

Quantum Mechanics, Spectroscopy, and Statistical Thermodynamics

INSTRUCTOR: Sara E. Murphy (she/her/hers)

Email: smurphy4@oxy.edu

OFFICE HOURS: Wednesdays, 11:00 AM – 12:00 PM

Fridays, 1:00 PM – 3:00 PM

If you would like to set up another time to meet, feel free to email me.

TEXT: *Physical Chemistry* by Atkins, De Paula, & Keeler, 12th edition (Volume 2: Quantum Chemistry, Spectroscopy, and Statistical Thermodynamics, ISBN: 9780198851318). Additional readings will be posted on the Moodle course page.

LECTURES: MWF 08:30 AM – 09:25 AM, Mosher 3. Please be on time and prepared for lecture.

Course Description

This course provides an introduction to quantum mechanics, atomic and molecular structure, spectroscopy, statistical thermodynamics, and chemical applications of group theory.

Course Learning Outcomes

1. Understand the fundamental postulates of quantum mechanics and apply these principles to solve the Schrödinger equation for a variety of model systems.
2. Implement molecular orbital theory and approximation methods to model multi-electron atoms and the formation of chemical bonds.
3. Apply the quantum mechanical framework of molecular structure and group theory to understand the interactions between electromagnetic radiation and matter and interpret spectroscopic phenomena.
4. Understand the Boltzmann equation and apply it to relate microscopic and macroscopic thermodynamic quantities.

Assessments

1. *Problem Sets*

There will be **four** problem sets throughout the course. They will be posted on the Moodle site, and there will be approximately two weeks to complete each set. You are **encouraged** to work on problems in groups – however, each student must turn in their own solutions, and you must state your collaborators on the first page of your solutions. Solutions will be due at 11:59 PM on the due date as a pdf uploaded to Moodle. The final problem set will be longer than the first three and will stand in place of a final (you may still work collaboratively). **Please come to office hours – I love to work through problems with students and answer questions.**

2. *Exams*

There will be **two** exams during the course. Both will be take-home exams. You may not work with other students on these exams, but you may ask me questions and both exams will be open note and open book. Integral tables and mathematical resources will be provided. You will have 48 hours to complete the exam.

Tentative Course Schedule

This schedule is subject to change.

Dates	Topic	Exams and Assignments	Readings
1/22 – 1/26	Quantum Theory (Learning Outcome #1)		Focus 7A-7F, 8A
1/29 – 2/2		Problem Set 1 Assigned	
2/5 – 2/9			
2/12 – 2/16		Problem Set 2 Assigned	
2/19	No Class		
2/21 – 2/23	Molecular Structure (Learning Outcome # 2)		Focus 8B, 9A-9E
2/26 – 3/1			
3/4 – 3/8		Exam # 1	
3/11 – 3/15	No Class – Enjoy Spring Break		
3/18 – 3/22	Spectroscopy, Symmetry, and Group Theory (Learning Outcome # 3)	Problem Set 3 Assigned	Focus 8C, 10A-10C, 11A-11G
3/25 – 3/29			
4/1 – 4/5			
4/8 – 4/12		Exam # 2	
4/15 – 4/19	The Boltzmann Distribution (Learning Outcome # 4)	Problem Set 4 Assigned	Focus 13A
4/22 – 4/26			
4/29	Last Day of Class		

Grade Breakdown

Percentage	Grade
92% - 100%	A
89% - 91.9%	A-
87% - 88.9%	B+
82% - 86.9%	B
79% - 81.9%	B-
77% - 78.9%	C+
72% - 76.9%	C
69% - 71.9%	C-
67% - 68.9%	D+
62% - 66.9%	D
59% - 61.9%	D -
58.9% or below	F

Problem Sets – 50%

Problem sets *must* be turned in on the due date for full credit. For each week that a problem set is late, 10% will be removed from your grade. If you need an extension, please email me *at least* 24 hours in advance of the due date. You will be able to drop one of the first three problem sets from your grade.

Exams – 40%

Participation – 10%

Attendance at lecture is required to keep up with the course material. At the end of each lecture, you will be asked to submit your “muddiest point” from that lecture (a brief note of the thing that confused you the most from that lecture). Your participation grade will be determined by your rate of lecture attendance and your submission of your muddiest point at the end of each lecture. However, I am aware that life happens - if you need to miss a lecture for personal reasons, illness, or other concern, please let me know *via* email.

Diversity and Inclusion

All students, regardless of their identity, are entitled to a positive and inclusive learning environment – disrespect, discrimination, and harassment will not be tolerated. If you have any concerns with the classroom learning environment, feel free to speak to me. Any concerns or suggestions for creating a more inclusive classroom will be given my prompt attention and serious consideration.

Academic Ethics and Integrity

Please see the student handbook at <https://www.oxy.edu/student-handbook> for information on school policy surrounding academic ethics.

Accommodation Requests

Students with documented disabilities who are registered with Disability Services should present their accommodation letter to me at the beginning of the semester or as soon as possible thereafter. Any student who experiences a physical or mental health impairment is encouraged to contact Disability Services at (323) 259-2969 to learn about available services, accommodations, and support. More information is available at <https://www.oxy.edu/offices-services/disability-services>.

Mental Health and Wellbeing

College can be a stressful and tumultuous experience, especially if you are experiencing mental illness and/or stressful life circumstances. If you are in distress, please know that help is available and visit the [Emmons Wellness Center](#) for mental health support.

If you are in extreme distress and are considering hurting yourself, please know that you are not alone and there are resources available. Please call the National Suicide Prevention Hotline at 1-800-273-TALK (8255).

Accommodations for Reasons of Faith and Conscience

Consistent with Occidental College's commitment to creating an academic community that is respectful of and welcoming to persons of differing backgrounds, we believe that students should be excused from class for reasons of faith and conscience without academic consequence. While it is not feasible to schedule coursework around all days of conviction for a class as a whole, faculty will honor requests from individual students to reschedule coursework or be absent from classes that conflict with the identified days. Information about this process is available on the ORSL website: <https://www.oxy.edu/student-life/resources-support/orsl>.

Title IX and Sexual Harassment

Please review the college's [Title IX Policy](#), including the statement indicating the role of faculty members as mandatory reporters. All faculty members are mandatory reporters in any instance of sexual misconduct. I **cannot** keep information about sexual misconduct confidential if you share this information with me. In the event that you choose to write or speak to me about having experienced sexual or interpersonal violence, including sexual assault, dating violence, domestic violence, stalking, sexual exploitation, or any form of sexual harassment, federal and state education laws require that I, as your instructor and a designated Responsible Employee, notify the Title IX office. They will contact you to let you know about accommodations and support services at Oxy and reporting options both on- and off-campus. For a list of **confidential resources** both on- and off-campus, please see <https://www.oxy.edu/civil-rights-title-ix/sexual-respect-title-ix/resources>.